The State of ZK

ZKV

Q4-2023

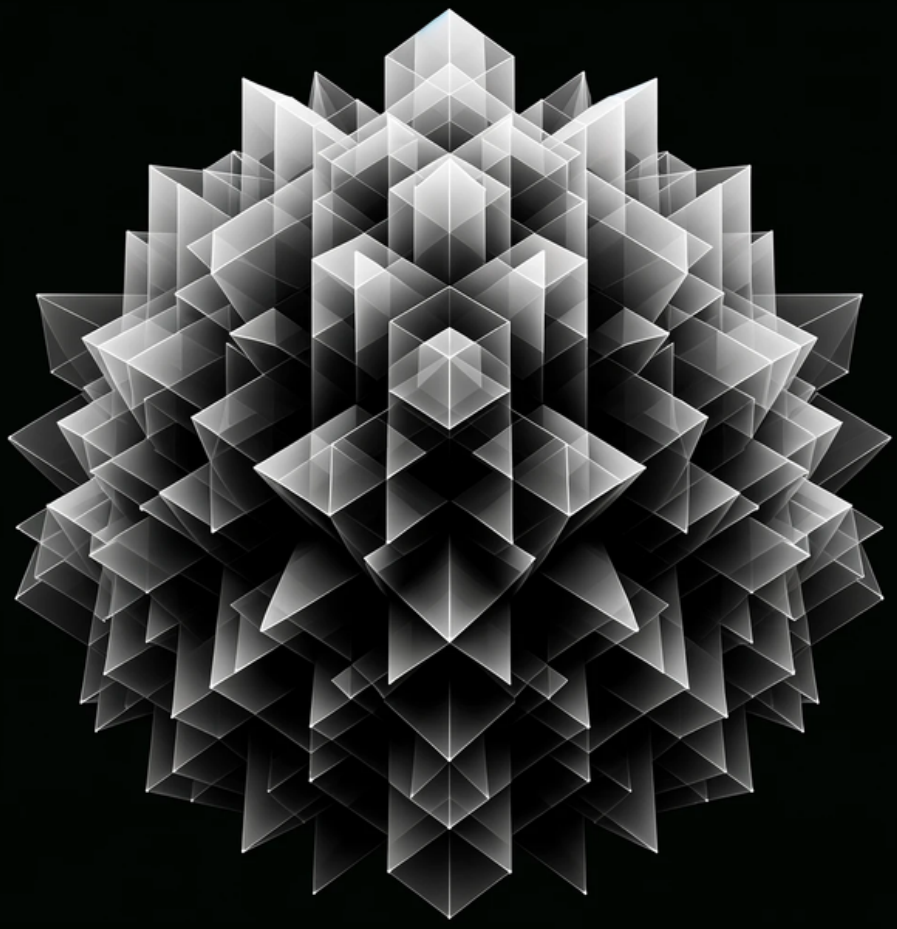


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Introduction

We're excited to share with you the latest edition of the State of ZK Report for Q4 2023. In this report, we recap what has happened in the ZK ecosystem during the final quarter of the year and highlight the key topics and takeaways.

In Q4, we saw the release of some innovative research, but like much of 2023, the majority of the ecosystem was focused on implementation and engineering. More and more long-standing ZK projects edged towards a new testnet or network release and many of the ZK frameworks became ever more usable. Meanwhile, some novel ZK-powered applications began to see some early and promising adoption. In this report, we dive into ZK for ID, one such application which is starting to see glimmers of real end-user adoption. The products that we will cover here are zPass, Polygon ID, ZK Email, ZKP2P (which was built on ZK Email) and ZK Login.

As an industry, we continue to struggle to find clear metrics on which to gauge & compare ZK proving system. However, this past quarter, researchers from both academia and industry are taking on the challenge and developing frameworks in which to better analyse and compare these projects.

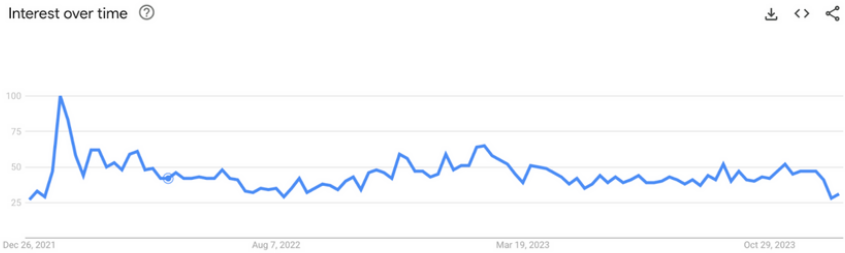
On the investment front, total volume of venture investment was down, with fewer and smaller ZK raises than previous quarters. Still, developer interest in ZK remains high and growing with more developers joining ZK communities and teams. The quality of the teams joining the ZK space and the sophistication of some of the emerging ZK applications offers us a strong indication that 2024 could be the ZK breakthrough year!

A bit about us: ZKV is a mission-driven company running validator and node infrastructure on over 10 networks. We were founded in 2019 and our team has strong ties to many of the organizations shaping the ZK industry, such as The Zero Knowledge Podcast, ZK Hack, zkSummit, rhino.fi, Geometry Research, and UCL. We have led multiple ZK-focused initiatives including public goods funding, events series and legal education. You can find out more about our work at www.zkvalidator.com.com.

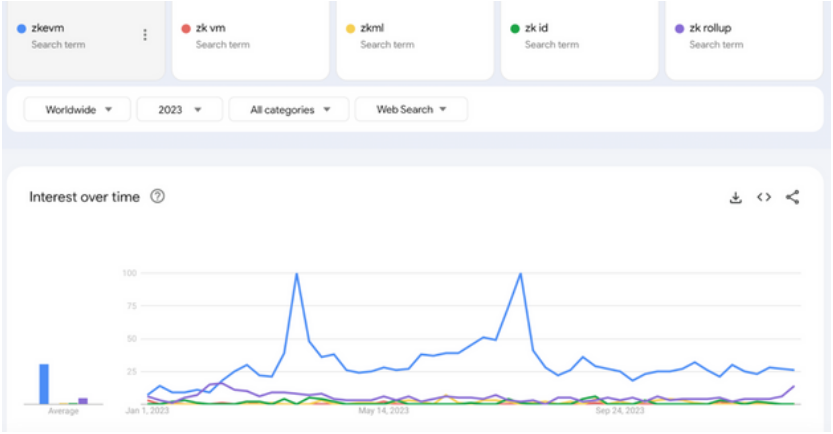
ZK in Numbers

ZK Search intents on Google 2022 vs. 2023

Last year, searches on Google for “Zero Knowledge Proofs” were mostly steady through 2023 compared to 2022, experiencing a peak during the week of February 12th. During 2022, searches were primarily about the concept around ZKPs. In 2023, we saw more searches for specialized terms such as zkEVM, ZK Rollups, etc. Starting in September, the interest rose slightly reaching a seasonal peak in early October. According to this data, some of the “rising” topics for the year include Polygon zkEVM and zkML.



When analyzing other ZK terms, we found that “zkEVM” was the most popular search. ZK Rollup saw a small uptick at the end of the year. zkVM, ZKML and ZKID remain niche but are picking up some interest on Google.

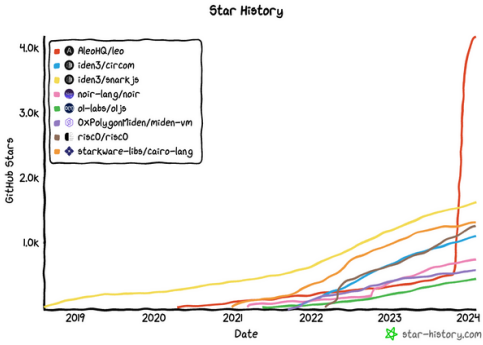


ZK in Numbers

Benchmarking DSLs

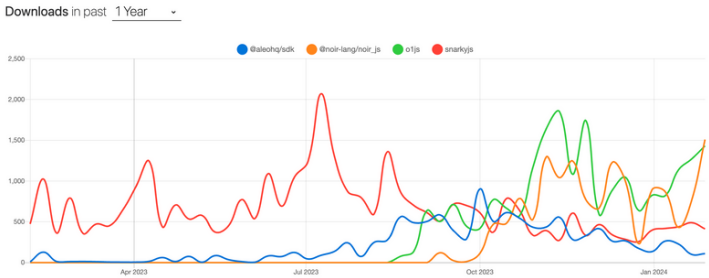
We would like to follow up on the survey results about zkDSLs that we presented in the previous Q3 edition of the State of ZK report. For this Q4 edition, we examined the data on the downloads, stars and usage of each of the ZK DSLs.

Here are the findings:



Github Stars: On January 11th, Aleo's Leo had the highest number of stars on GitHub with 4453. SnarkJS followed with 1590 stars, Cairo with 1305, Circom with 1063, Noir with 725, and o1js with 426. It's worth noting that Leo's leadership has grown significantly since October when it had only 540 stars and was second to last on the list. Note that all the other DSLs have remained in the same ranking as stated since November 2021.

Downloads: According to npmtrends, SnarkJS leads in downloads, followed by Circom, Noir, and o1js.



In conclusion, although there is a clear rising interest in some of the most recent DSLs or JS libraries such as Leo and Noir, most of the activity continues to happen with older frameworks like Circom, and Cairo.

Source: https://npmtrends.com/@aleohq/sdk-vs-@noir-lang/noir_js-vs-o1js-vs-snarkyjs
https://npmtrends.com/snarkjs-vs-starknet
https://star-history.com/#AleoHQ/leo&iden3/circom&iden3/snarkjs&noir-lang/noir&o1-labs/o1js&0xPolygonMiden/miden-vm&risc0/risc0&starkware-libs/cairo-lang&Date

ZK in Numbers

ZK protocols by the numbers

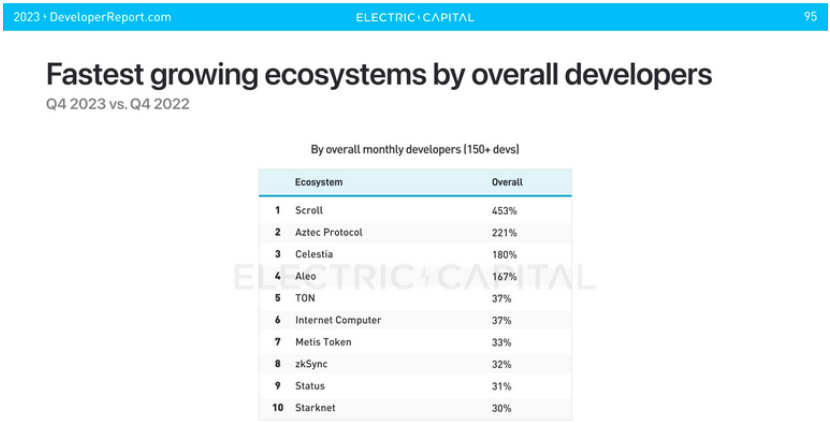
The Electric Capital report shed some light on the state of ZK protocols. Here are some of the findings according to the report:

- In terms of new developer onboarding, Starknet was the leading ZK rollup ecosystem, concluding the year with 1,291 new developers, ranking it in the Top 15 within the general blockchain ecosystem. Within the ZK Rollup field, it was followed by zkSync, Scroll, and Polygon zkEVM in developer growth.

- zkSync and Starknet are distinguished as the only two ZK Rollup ecosystems with over 1,000 active developers.

- In terms of overall developer acquisition, five out of the ten fastest-growing blockchain ecosystems are ZK-based. Scroll leads with 453% year-over-year growth, followed by Aztec Protocol with a 221% increase in 2023. Other notable protocols include Aleo (167%), zkSync (32%), and Starknet (30%).

- Regarding the growth of full-time developers, five of the top 10 networks are ZK ecosystems. Here, Aztec Protocol stands out as the fastest-growing network with a 279% increase, followed by Mina Protocol, Scroll, zkSync, and Starknet.



Breakthrough research

zkBench: zkBench is the first open-source, generic, extensible, and configurable benchmarking framework designed for public-key cryptography and ZKPs. It offers extensible backends for benchmarking both low-level arithmetic operations and high-level ZKP circuits, using exemplary test vectors such as SHA-256 implementations in various ZKP tools or libraries. zkBench conducts comprehensive benchmarking of high-level ZKP circuits, covering execution time and memory consumption across the setup, proving, and verification phases. Additionally, it introduces a novel methodology that enables developers and researchers to perform detailed performance analysis of cryptographic algorithms and protocols.

Cicada: Cicada is a general framework designed to facilitate practical, privacy-preserving, and trust-minimized protocols for both auctions and voting. It achieves its objectives through the use of time-lock puzzles (TLPs). These TLPs function as commitments to bids or ballots, which can be opened by any party after a predefined period. This approach ensures privacy and reduces the need for interactivity in a trustless environment, notably by eliminating the requirement for a second reveal round.

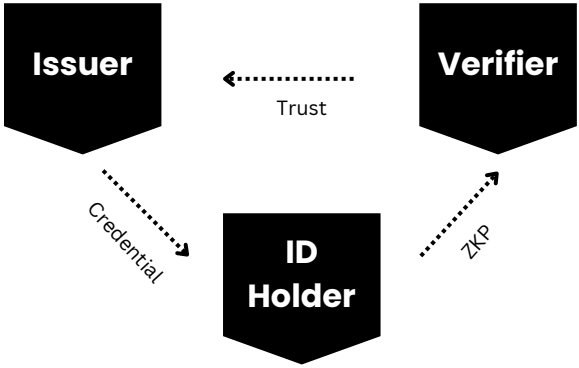
BabySpartan: Lasso-Based SNARK for Non-Uniform Computation: BabySpartan is an advancement in succinct non-interactive arguments of knowledge (SNARKs), merging SuperSpartan and Lasso protocols for Plonkish constraint systems. It focuses on improving computational efficiency by reducing cryptographic commitments to small values, addressing a challenge in SNARK technology deployment. Integrating SuperSpartan's framework with Lasso's commitment approach, BabySpartan aims to enhance the application of SNARKs in blockchain technologies. This development highlights a move towards specialized solutions within the blockchain industry, aiming to address technical challenges and expand blockchain capabilities.

Binius: The Ulvetanna team's "Succinct Arguments over Towers of Binary Fields" introduces Binius, a novel SNARK construction approach using binary fields (GF(2)). This method aligns closely with digital circuit data types, significantly reducing memory and computational overhead. Binius features a polynomial commitment scheme for GF(2) with no embedding overhead and supports a range of binary field sizes.

State of the Art in ZK ID

In our theme of this edition, we dive into the topic of ZK ID. At the core of all the ZK ID systems, ZKPs are used to ensure trust and privacy as well as prove the validity and correctness of underlying data. These systems are all built around three parties:

- **The issuer:** the party who issues the credentials. It could be a government agency, a smart contract or any other organization.
- **The identity holder or prover:** in this case, this is the user trying to prove to the verifier that they meets certain criteria using the credentials issued by the issuer.
- **The verifier:** this is the party interested in knowing if the characteristics claimed by the prover are true in order to execute an action (i.e., provide the prover with a service only available to certain populations).



This particular system is not new. It shares some similarities with the way Single Sign-on (SSO) logins function on the Internet. In that case, to authenticate oneself with a service provider, the user must present proof of their identity using credentials provided by Google or Facebook (the issuers), while the service provider acts as the verifier. However, the key difference between this model and ZK ID is in the way the parties involved interact with each other, and how data is managed and processed.

In most ZK ID systems, the credentials are owned by the ID Holder and stored in their wallet. This allows the ID Holder to have complete control over their identity. Furthermore, when the ID Holder is proving their identity to a verifier, the privacy-preserving capabilities of ZKPs ensure that no information is shared. Instead, the proof simply confirms the validity of the credential.

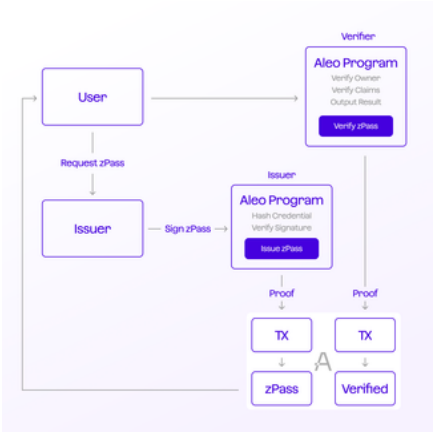
State of the Art in ZK ID

Main Players

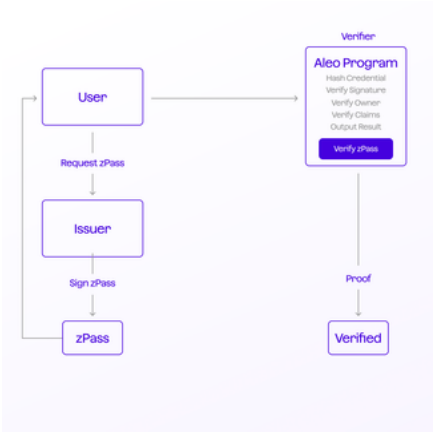
As stated in the introduction, this section covers some of the projects in the ZK ID space. Here's an overview of how each of them works:

zPass integrates ZKPs, specifically zk-SNARKs, to enhance the privacy and programmability of identity verification, using Aleo's infrastructure for security and scalability. Aleo facilitates this through its implementation of ZEXE, enabling encrypted data storage and private transactions while preserving user anonymity. The system adopts a record model inspired by Zcash's UTXO model, which supports complex operations with private inputs and outputs, ensuring data remains encrypted and user-controlled. By decentralizing computation and employing off-chain execution in conjunction with ZKPs, zPass ensures that verification does not compromise data privacy.

Onchain Workflow

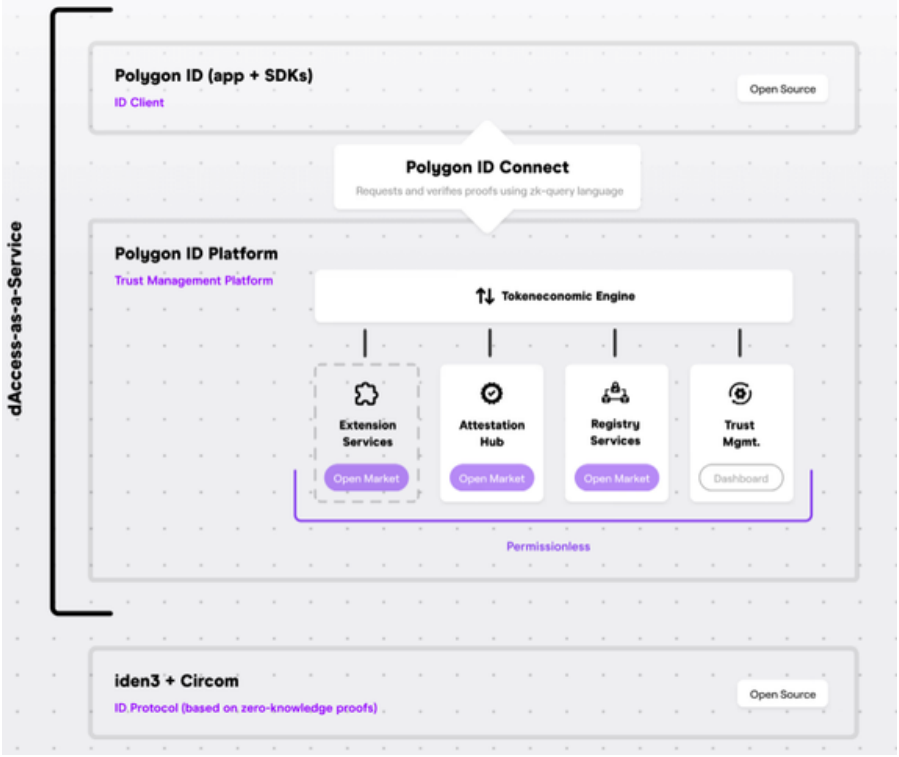


Offchain Workflow



State of the Art in ZK ID

Polygon ID's architecture integrates on-chain and off-chain elements alongside zero-knowledge proofs to construct a privacy-focused identity system. Its foundation lies in an identity state commitment on the blockchain, organized into three Merkle trees: the claims tree for individual claims storage, the revocations tree for logging invalidated claims, and the roots tree as a chronological record of identity states. This structure allows for claim retrieval, verification, and revocation. Claim interactions involve issuers adding claims with digital signatures to the claims tree, and users presenting these claims along with Merkle tree proofs to verifiers, who can ascertain the claim's validity and non-revocation status without exposing sensitive data.

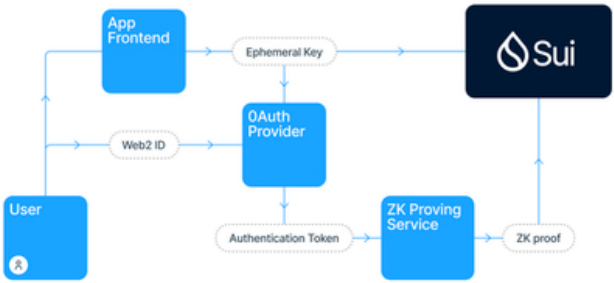


State of the Art in ZK ID

A new generation of ZK ID project bridge the gap between Web2 IDs and Web3 protocols have emerged. Here are a few:

ZK Email: a protocol that leverages DomainKeys Identified Mail (DKIM) signatures, which are cryptographic signatures attached to emails that verify the sender's domain, to enable users to prove they received an email from a specific sender. This method allows for the verification of off-chain data, such as email-based attestations, in a decentralized and privacy-preserving manner, leveraging ZKPs to maintain user anonymity while providing specific information.This can be effectively used to connect Web2 identities to onchain addresses.

zkLogin: similar in its architecture to ZK Email, zkLogin is a Sui primitive that provides the ability for users to send transactions from a Sui address using an OAuth credential, without publicly linking the two.

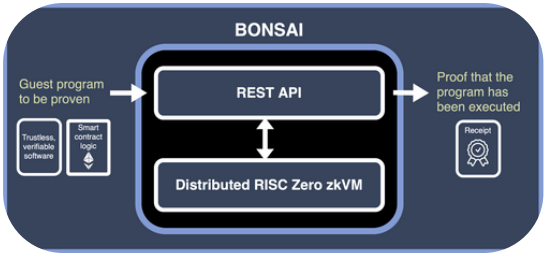


ZKP2P: a trustless Fiat Onramp built on top of Venmo using ZK Email to confirm the authenticity of transactions without disclosing sensitive information. *Fun fact: the original ZKP2P project was built at the ZK Hack Lisbon Hackathon. ZK Hack is a project that ZKV helped initiate.*

On a sad note, Sismo, one of the "rising stars" in the ZK ID industry, had to shut down in Q4 due to it being "unsustainable" for the small team, as stated by its founder. The team has since open-sourced the code they have built, and we hope that someone will continue to build these tools as the idea remains compelling as an ID solution - that is a system that uses on-chain activity to help build a reputation (and identity).

Product launches

Project	Launch
Succinct	Succinct launched two important partnerships: one with Avail to develop Vector X and a second one with Celestia named Blobstream. Vector X is a ZK light client designed to secure Avail's data attestation bridge to Ethereum, utilizing zkSNARKs for verifying Avail's GRANDPA consensus. This enables efficient, on-chain light clients that store Avail's data roots and state roots, enhancing Ethereum-settled rollups' access to Avail's data attestations. Similarly, BlobstreamX, created in collaboration with Celestia, relays data commitments to an Ethereum on-chain light client, supporting Ethereum-settled L2s through efficient data verification. Both Vector X and BlobstreamX use ZK proofs to ensure cost-effective and trust-minimized interoperability.
RiscZero	Bonsai Pay was launched to facilitate entry into the Ethereum network using existing Google accounts and other major identity providers via OpenID Connect. Developed with RISC Zero's zkVM and Bonsai proving service, it offers a straightforward way for users to interact with the Ethereum ecosystem. The application simplifies Ethereum transactions and withdrawals, linking online identities to blockchain actions securely through ZKPs.

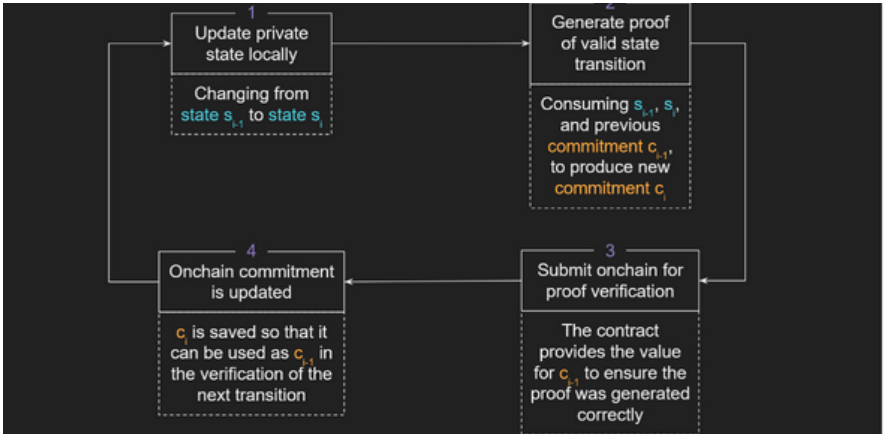


Product launches

Project	Launch
Lurk	Arecibo, a fork of Nova, recently got its first crate released, positioning itself as an incubation project for Nova. It focuses on testing and improving Nova's folding framework, with the introduction of SuperNova, a protocol designed to optimize proof statements and reduce computational complexity. Aimed at rapid evolution, Arecibo offers limited backward compatibility and plays a vital role in supporting the Lurk release. This initiative is a collaborative effort, with significant contributions aimed at enhancing cryptographic proofs and folding schemes, and ultimately contributing back to the Nova project.
Nocturne	Nocturne v1, a protocol working on private and composable accounts on Ethereum, launched in November, experiencing a steady increase in deposits and user operations. In response to initial feedback, efforts are underway to enhance error communication and offchain reliability, particularly in handling private balance gas payments and transaction submission during gas price fluctuations. Future developments for Nocturne focus on reducing gas costs via Layer 2 solutions, enhancing permissionlessness through proof of innocence, and facilitating wallet authorization reuse.
Aztec	The Aztec team released Noir Beta. This latest version enhances stability and integrates seamlessly with Aztec's UltraPlonk proving backend, also featuring browser compatibility through NoirJS. While still unaudited, Noir now offers a mature development experience and a comprehensive feature set, making it more accessible for developers.

Initiatives

ZK Hunt by Flynn Calcutt: 0xParc published ZK Hunt, a videogame developed during the 0xPARC 2022 Autonomous Worlds Residency. It is an onchain PvP RTS game using innovative ZK mechanics and information asymmetry. Leveraging the MUD framework and circom for contract architecture and ZK proof generation, the game evolved from a concept of private onchain movement to a platform showcasing diverse cryptographic mechanics, ultimately serving as an educational tool in ZK game development.



zkUniswap by Diego Estevez (0xfuturistic): zkUniswap is a conceptual fork of UniswapV3 utilizing a zkVM (RISCZero) for off-chain computation of swap steps. It processes swap requests made on-chain; a relay computes this off-chain, posting the output and proof back to the EVM. If validated, the swap executes. zkAMMs, like zkUniswap, integrate zero knowledge proofs into Automated Market Makers, offloading computation to zk coprocessors. This approach, cheaper than on-chain computations, maintains trust guarantees through on-chain proof verification. zkAMMs represent an evolution in decentralized trading platforms, enhancing efficiency while retaining the security and trust of blockchain technology. Additionally, they have the potential to offer differential privacy in transactions, a feature that enhances user privacy by adding noise to data, preventing individual data points from being distinguished.

Initiatives

ZK Score by Omer Shlomovits (Ingonyama), Lucy Williams: ZK Score is a proposed metric system for evaluating Zero Knowledge Proof (ZKP) technologies, focusing on efficiency at the hardware level. It measures the number of proofs generated per joule of energy consumed, providing a standardized way to compare ZK tech stacks. The goal of ZK Score is to offer a tangible, hardware-centric benchmark, akin to those in AI technologies, to assess the performance of various ZKP implementations. By measuring proofs-per-Joule, ZK Score aims to quantify the energy efficiency of ZKP systems, enabling a fair and objective comparison.

Aleo Testnet 3 Phase III results: The Aleo Deploy Incentives program during Testnet 3 Phase III concluded with over a thousand submissions, leading to the selection of the top 10 winners sharing a 5M Aleo Credits prize pool. Key learnings from the program included insights on enhancing the Leo programming language and refining Aleo's core protocol, informed by practical feedback and real-world application testing. These learnings are instrumental in guiding Aleo's ongoing development and optimization efforts.

Halo2-repl by Axiom: The release of halo2-repl by Axiom, a browser-based REPL for halo2 in JavaScript, aims to simplify the creation of ZK circuits for developers. It eliminates the need to learn new programming languages and manage complex installations, enabling direct in-browser compilation and proof generation of ZK circuits. This tool utilizes WebAssembly bindings for integrating Rust's ZK circuit primitives into a JavaScript environment, fostering easier access and experimentation with ZKPs in blockchain technology.

Penumbra Summoning Ceremony: The Penumbra Summoning Ceremony's Phase 1 is complete with over 11,000 participants. The team announced a brief pause before Phase 2.

Namada crossed 100K: After announcing its airdrop, Namada has achieved another milestone by reaching over 100,000 downloads of its Google Chrome extension. The project is currently running an incentivized testnet in preparation for the mainnet launch.

Friction points

01 | **Aztec Claim Proof Bug**

On September 12th, a bug was reported in Aztec Connect's ZK circuit, identified by security researcher lucash-dev via their Immunefi bug bounty program. Aztec quickly responded, disabling vulnerable workflows and patching the system by October 3rd. The bug affected the claim proof in their privacy-focused zkRollup system, allowing sequencer exploitation due to complexities in finite field arithmetic used in ZK circuits, necessitating intricate integer operation solutions.

02 | **zkSync ZK Circuits**

On September 15th, ChainLight researchers identified a soundness bug in the ZK-circuits of the zkSync Era mainnet, promptly reporting it on the 19th. The bug permitted a malicious prover to generate false "proofs" for improperly executed blocks, which would be accepted by the L1 verifier smart contracts. Matter Labs responded by deploying a fix. The bug revolved around the handling of memory write instructions in the EraVM, allowing the alteration of transaction values.

Ecosystem Investment Rounds

Here were the public fundraising rounds in the ZK industry during Q4.

Project	Industry	Amount
Nocturne	Privacy	$6M
Modulus	ZK for IA	$6.3M
Union Labs	Interoperability	$4M

Conclusion

In Q4 2023, investments in ZK were down but developer interest was up. This can be attributed at least in part to the imminent launch of Aleo, Namada, and Penumbra as well as Mina's zkApps, Aztec Sandbox, and the RISC Zero General-Purpose Verifiable Computing environment. On the application front, the new generation of ZK ID projects demonstrated new ways to port existing web2 & offchain identities into web3. This is excellent news for both the ecosystem and the industry as a whole. We are excited to see what 2024 has in store for ZK!

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